

Math 107-01  
Test 4A: December 4, 2013

Answer all of the following questions. You will have 50 minutes to complete the test. You may not use any outside assistance on this test. You may use a calculator, but the use of electronic equipment such as mp3 players, ipods, cell phones and any electronic devices other than your calculator during the test is prohibited. If you have time, check your answers for accuracy.

**Be sure to show all work. Unsupported answers may not receive credit.**

Good luck!

Name: \_\_\_\_\_

1. The data below shows the number of credit hours the members of the clogging team are taking this semester. Construct a frequency distribution (5 points) and histogram (7 points) to represent the data. Use 5 subintervals, beginning at 12 and ending at 22.

|    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|
| 12 | 12 | 12 | 12 | 12 | 13 | 13 | 13 | 13 | 13 |
| 13 | 13 | 14 | 14 | 14 | 14 | 14 | 15 | 15 | 15 |
| 15 | 15 | 15 | 15 | 15 | 15 | 15 | 15 | 15 | 16 |
| 16 | 16 | 16 | 16 | 16 | 17 | 17 | 17 | 17 | 17 |
| 17 | 17 | 18 | 18 | 18 | 19 | 19 | 20 | 20 | 21 |

2. A group of 10 MHU students were randomly selected and asked how many parking tickets they have received on campus this semester. The following data was recorded:

0      0      1      1      1      2      2      4      5      7

- (a) (4 points) Find the mode of this data set.

- (b) (5 points) Find the mean of this data set.

- (c) (4 points) Find the median of this data set.

- (d) Find the variance (5 points) and standard deviation (2 points) of this data set. Round answers to two decimal places.

[illegible]

3. The weights of boxes of cereal are normally distributed with a mean weight of 18.0 ounces and a standard deviation of 0.2 ounces.

(a) (5 points) Find the interval that defines one standard deviation of distance from the mean. Approximately how much data fall within this interval?

(b) (5 points) Find the interval that defines two standard deviations of distance from the mean. Approximately how much data fall within this interval?

(c) (5 points) Find the interval that defines three standard deviations of distance from the mean. Approximately how much data fall within this interval?

4. Shade the area under the bell curve for each probability (2 points). Circle the answer choice for the probability represented (2 points each).

(a)  $p(-1.25 < z < 1.78)$

a) 0.3944

b) 0.0681

c) 0.8569

d) 0.4625

(b)  $p(z > -1.14)$

a) 0.3729

b) 0.8729

c) 0.4922

d) 0.1271

(c)  $p(z > 2.31)$

a) 0.3888

b) 0.4896

c) 0.9896

d) 0.0104

(d)  $p(z < -0.54)$

a) 0.2054

b) 0.2946

c) 0.7054

d) 0.3830

(e)  $p(0.81 < z < 2.14)$

a) 0.1928

b) 0.2910

c) 0.4838

d) 0.7748

5. (7 points) The amount of time between taking a pain reliever and getting relief is normally distributed with a mean of 27 minutes and a standard deviation of 6 minutes. Find the probability that the time between taking the medication and getting relief is between 24 and 36 minutes. You must show all work to receive full credit.

6. The weight (in grams) of a can of Friendly Fish tuna is normally distributed with a mean of 185.7 grams and a standard deviation of 0.96 grams. You must show all work to receive full credit.

- (a) (6 points) What is the probability that a can of tuna weighs less than 184 grams?

(b) (6 points) What is the probability that a can of tuna weighs more than 188 grams?

(c) (6 points) What is the probability that a can of tuna weighs between 183 grams and 187 grams?

7. (8 points) Professor Jones assumes that Math 107 Test 4 scores are normally distributed. The mean score was 65, with a standard deviation of 9. If Professor Jones wants the top 15% of the students to receive an A, find the minimum score to receive an A. You must show all work to receive full credit.

On my honor, I have neither given nor received any academic aid or information that would violate the Honor Code of Mars Hill University. \_\_\_\_\_(please sign)